1 AI Governance — framing

AI Governance = AI Regulation = 'Legalizing AI' (used interchangeably). Core open question: **is AI itself governed, or are its USES governed?** And can one policy work across **sectors** (aviation, agriculture, education), **geographies**, and **tasks** (facial recognition, summarization…)?

Countermeasures (from computer security), applied to AI: (1) train the users, (2) UI-level warnings, (3) **silently eliminating the threat**.

Threat-mitigation styles (Q4): when a system (e.g. an email spam filter) removes malicious content **in the background without user intervention**, it is **silently eliminating the threat** — not educating users, not interface warnings, not enforcing legal rules.

Right to be Forgotten (Q1): GDPR **Article 17** — the right that **requires companies to delete user data on request** (vs data portability, transparency, or IP retention).

2 OECD AI Principles

Ten value-based principles for trustworthy AI (two groups):

Values-based principles:

- Inclusive growth, sustainable development & well-being
- Human rights & democratic values, incl. fairness & privacy
- Transparency & explainability
- **Robustness, security & safety**
- Accountability

Recommendations to policymakers:

- Investing in AI research & development
- Fostering an inclusive AI-enabling ecosystem
- Shaping an enabling governance & policy environment
- Building human capacity / preparing for labour-market transition
- International co-operation for trustworthy AI

Assignment: 'AI should operate safely & function properly under normal use, misuse, or adverse conditions' = **Robustness, security & safety** (Q2). Under **Accountability**, ensuring **traceability** means one can **trace back how data was collected & how it influences decision-making** (Q7).

3 The EU AI Act — risk-based, horizontal regulation

First comprehensive AI law. Parliament's priority: AI in the EU must be **safe, transparent, traceable, non-discriminatory & environmentally friendly**, and **overseen by people, not automation**. Four risk tiers:

Tier	Rule / examples
Unacceptable (BANNED)	**Cognitive behavioural manipulation**; **Social scoring**; biometric categorisation; real-time remote biometric ID (facial recognition) — limited law-enforcement exceptions
High-risk	Negatively affect safety/rights: product-safety areas (toys, aviation, cars, medical devices) + critical infrastructure, education, employment, law enforcement, migration — assessed before & during market
Limited-risk (transparency)	Generative AI (e.g. ChatGPT): disclose content is AI-generated; prevent illegal content; publish copyrighted-data summaries; **deepfakes must be clearly labelled AI-generated**
Minimal-risk	Free use

Assignment: unacceptable-risk (banned) = **Social Scoring Systems + Cognitive Behavioural Manipulation** (Q3; critical-infra & law-enforcement are high-risk, not banned). Deepfakes/generated media must be **clearly labelled as AI-generated** (Q5).

4 Who's Who under the EU AI Act (the value chain)

Actor	Role
Provider	Develops AI / GPAI models for release under its own name (free or commercial)
Deployer	**Persons/orgs that USE AI systems under their own authority**
Manufacturer	Provides/distributes AI in products under its own name/trademark
Importer	EU person releasing AI bearing a non-EU provider's name
Distributor	Makes AI available in the EU market
Representative	EU person appointed by provider to perform AI-Act obligations

Assignment Q6: the **'deployer'** = the **person or organization that uses the AI system under its own authority**. (GPAI = General-Purpose AI.) **AI Safety Institutes (AISI)** exist in the UK, US (NIST), EU & India's National AI Mission.

5 SaGE — Quantifying Moral Consistency in LLMs

Semantic Consistency (Elazar 2021) = the ability to make **consistent decisions in semantically equivalent contexts**: semantically equivalent questions should yield semantically equivalent answers. **Claim: LLMs are NOT semantically consistent** — they give contradictory answers to paraphrased questions.

Why it matters: a model can look fine in testing but **behave unexpectedly when deployed**, hurting trust & reliability (a safety risk). Worse in **moral scenarios** (conflicting training data) than commonsense ones.

Semantic inconsistency (Q8): providing semantically disparate answers to semantically equivalent prompts.

Semantic Graph Entropy (Q9): less entropy = consistent, MORE entropy = inconsistent. So a **higher** semantic graph entropy indicates the model is **more semantically INCONSISTENT**. (Built from an information functional over the graph; Dehmer & Mowshowitz 2011.)

Temperature (Q10): is consistency dependent on temperature? **Probably not** — lowering temperature leaves the consistency score **largely unaffected.**

The SaGE pipeline:

- **Paraphrase generation** — few-shot Vicuna-13B; filter with ParaScore >0.8 (high semantic similarity + lexical divergence)
- **Rule of Thumb (RoT)** generation — a fundamental judgment of right/wrong per query+response
- **Moral Consistency Corpus (MCC)** — 10K questions, 5 paraphrases each, 11 LLMs
- **Semantic graph** — responses → SBERT/DeBERTa embeddings (fine-tuned on NLI); each response = a node; edge weight = cosine distance
- **Semantic Graph Entropy** — the consistency metric

Key findings:

- Current LLMs are **very inconsistent** (best score only ~0.575)
- **Accuracy on benchmarks does NOT correlate with consistency** (Q11) — they are **separate problems**; evaluate consistency on its own
- **RoT prompting improves** consistency (GPT-3.5: 0.438 → 0.548) → hints RAG-style methods could help

SaGE quick-key: Q8 = semantically disparate answers to equivalent prompts; Q9 = higher entropy → higher inconsistency; Q10 = temperature leaves consistency largely unaffected; Q11 = accuracy does NOT correlate with consistency.